FixMyCity: A Location-Aware Citizen Grievance Redressal and Municipal Infrastructure Management Platform

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***Abstract*— Urban municipal infrastructure—such as roadways, street lighting, drainage networks, and solid waste collection—frequently experiences localized degradation that goes unnoticed due to fragmented grievance redressal channels. Traditional bureaucratic reporting mechanisms rely on manual paperwork, centralized call centers, or disjointed web portals, resulting in prolonged turnaround times, absence of visual evidence, and a lack of real-time progress transparency. This paper presents FixMyCity, a location-aware, cloud-native mobile and web platform designed to democratize urban governance and expedite municipal issue resolution. FixMyCity enables citizens to report public infrastructure defects using geolocation tagging, categorical filtering, photographic evidence, and a multi-stage status tracking lifecycle. The client application is constructed natively using modern Android Jetpack Compose, Kotlin Coroutines, and the Model-View-ViewModel (MVVM) architecture with reactive StateFlow streams. The back-end infrastructure combines Google Firebase (Firebase Authentication, Cloud Firestore NoSQL database, and Cloud Storage) with a local Node.js/Express REST server and administrative web dashboard. This dual-backend setup supports administrative triage, engineer assignment, and citizen notifications. A robust offline-first synchronization strategy guarantees full application responsiveness during network failures via local caching and graceful mock-data fallback. Experimental evaluations demonstrate instantaneous coordinate acquisition, sub-second reactive state updates, and deterministic report lifecycle progression, underscoring the platform's efficacy in fostering participatory urban governance.**

***Index Terms*—** Civic engagement, grievance redressal, smart cities, mobile crowdsensing, Android Jetpack Compose, Kotlin, Firebase, Cloud Firestore, location-based services, participatory governance.

I. INTRODUCTION

Rapid urbanization has placed unprecedented strain on municipal corporations tasked with maintaining public physical infrastructure. Potholes, broken streetlamps, overflowing garbage receptacles, damaged pedestrian walkways, and ruptured water mains degrade civic safety and public health. Despite substantial municipal budgets, maintenance authorities often suffer from severe "information lag"—critical defects remain unaddressed simply because municipal authorities lack real-time visibility into localized micro-level infrastructure failures.

Historically, citizens reported public grievances via localized ward offices, paper-based forms, or municipal telephone helplines. These legacy modalities suffer from persistent shortcomings: ambiguous spatial attribution, lack of photographic evidence, opaque resolution lifecycles, and a lack of unified analytics.

The widespread availability of GPS-equipped smartphones, ubiquitous wireless communication, and reactive mobile frameworks provides an ideal foundation for mobile crowdsensing and participatory urban governance [3], [4], [8]. By transforming citizens into active civic sensors, municipal corporations can crowd-source defect detection at negligible marginal cost [6], [10].

This paper presents **FixMyCity**, an end-to-end, location-aware civic grievance reporting and municipal resolution platform. Built natively for Android using Jetpack Compose and Kotlin Coroutines, the system integrates Firebase cloud infrastructure, interactive status progression timelines, role-based administrative dispatch, and offline-tolerant caching. FixMyCity bridges the trust deficit between urban citizens and municipal governance bodies by providing an open, accountable, and traceable issue-resolution ecosystem.

II. RELATED WORK

The concept of mobile crowdsourcing for public

infrastructure audit has gained significant academic and commercial interest. Pioneering initiatives like *FixMyStreet* in the United Kingdom established that web-based geolocation reporting substantially increases citizen participation in neighborhood upkeep [1], [2]. Similarly, Salovaara and Kankainen evaluated public engagement across digital municipal apps, concluding that ease of photo capture and immediate receipt confirmations directly govern citizen retention [2].

Mobile crowdsensing (MCS) paradigms, as surveyed by Ganti et al. [3] and Guo et al. [4], harness commodity smartphone hardware—such as accelerometers, cameras, and GPS receivers—to collect environmental and infrastructural data without specialized sensor deployments. Anthopoulos analyzed the integration of smart city architectures with public administration, highlighting that digital grievance pipelines must support both bottom-up citizen complaints and top-down municipal workflow dispatching [5]. Goodchild introduced the paradigm of Volunteered Geographic Information (VGI), demonstrating that geocoded user reports often match or exceed the positional accuracy of professional spatial surveys [8].

Recent engineering approaches focus on automating defect categorization and improving user interface responsiveness. Hiremath and Jadhav proposed an Android-based municipal complaint portal utilizing cloud databases; however, their architecture relied on older imperative UI toolkits, leading to state synchronization bottlenecks [11]. Castro et al. assessed the usability of municipal mobile applications and observed that multi-step reporting flows with non-responsive visual feedback cause significant user dropout [12]. Sinnott et al. evaluated urban incident reporting scalability and advocated for decoupled cloud persistence paired with client-side caching to sustain network brownouts [13]. In parallel, Diagne et al. applied smartphone telemetry to road defect reporting, emphasizing the necessity of GPS validation to eliminate duplicate or erroneous complaints [14].

A. Research Gap

Despite the proliferation of civic portals, existing platforms present substantial operational gaps: (1) absence of granular state timelines beyond binary "Open/Closed" states; (2) brittleness under intermittent connectivity during field reporting; and (3) decoupled administrative dashboards that fail to provide real-time bi-directional status synchronization. FixMyCity addresses these deficiencies through a modern declarative reactive UI, multi-stage state transitions, and dual-backend synchronization.

III. PROPOSED SYSTEM ARCHITECTURE

FixMyCity adopts a multi-tiered, event-driven architecture designed to balance rapid client-side rendering with robust distributed cloud persistence:

1) Client Presentation Layer (Android Application): Built natively using Android Jetpack Compose, Kotlin Coroutines, and the MVVM design pattern. State management is handled through unidirectional data flow using `StateFlow` and `SharedFlow`, ensuring reactive UI updates when underlying database states change.

2) Cloud Persistence and Identity Layer (Google Firebase): Firebase Authentication manages secure identity via email/password protocols and signed JSON Web Tokens (JWT). Cloud Firestore acts as the primary NoSQL operational document store, providing live snapshots and offline synchronization. Cloud Storage handles photographic evidence.

3) Administrative and Municipal Tier (Node.js/Express): Hosts a local REST API server alongside a Web Admin Dashboard on port 8080. This layer allows municipal supervisors to inspect incoming complaints, assign priorities, update statuses, and monitor operational performance metrics.

B. Functional System Modules

The platform decomposes into six cohesive modules: (1) Authentication Module for session management and credential caching; (2) Civic Complaint Reporting Module for defect categorization, reverse-geocoded location tagging, and image uploads; (3) Timeline and State Transition Engine supporting six discrete states (`SUBMITTED`, `UNDER_REVIEW`, `ASSIGNED`, `IN_PROGRESS`, `RESOLVED`, `REJECTED`); (4) Notification Engine dispatching live alerts; (5) Impact and Analytics Engine computing resolution percentages and citizen points; and (6) Offline-First Data Layer providing local caching and mock fallbacks.

IV. METHODOLOGY

The operational workflow of FixMyCity follows a closed-loop citizen-to-municipality interaction cycle. First, the citizen authenticates and captures a defect report with GPS coordinates and photographic evidence. The record is committed atomically to Cloud Firestore. Municipal officers inspect the incoming reports via the administrative dashboard and advance the workflow status. As work progresses, the citizen's mobile timeline updates reactively, triggering in-app alerts until final resolution.

A. Mathematical Formulation

To quantify municipal performance, the aggregate Municipal Resolution Ratio (η_res) across N total complaints is calculated as:

$$\eta_res = [\ \Sigma\ I(status_i = RESOLVED) / N\ ] \times 100\% \quad (1)$$

The Mean Time to Resolution (MTTR) across all resolved complaints R_res is expressed as:

$$MTTR = (1 / |R_res|)\ \Sigma\ (t_resolved - t_submitted) \quad (2)$$

Citizen civic reputation points (C_rep) are evaluated to incentivize community participation:

$$C_rep = \alpha \cdot N_sub + \beta \cdot N_res \quad (3)$$

where N_sub is the number of submitted reports, N_res is the resolved count, with weighting constants $\alpha = 10$ and $\beta = 25$.

V. SYSTEM IMPLEMENTATION

FixMyCity is implemented natively targeting Android SDK 35 (minSdk 26, Java 17). The presentation layer is built exclusively with Jetpack Compose and Material 3 design tokens. Asynchronous operations utilize Kotlin Coroutines across `Dispatchers.IO` and immutable `StateFlow` streams.

Image caching and asynchronous decoding are handled via Coil Compose. The repository layer (`FixMyCityRepository`) wraps Firestore network calls with structured exception handlers; when connectivity drops, operations gracefully fall back to cached local entities or bundled mock datasets, preventing app crashes.

VI. RESULTS AND DISCUSSION

The platform was tested across diverse network conditions and physical Android devices. Key functional test cases are summarized in Table I. The system demonstrated an average authentication latency of 410 ms, spatial accuracy within 5 meters, Firestore document sync latency of 320 ms, and notification delivery under 250 ms.

During intentional network disconnection, the offline-first caching mechanism allowed continuous report drafting and browsing with zero crashes, synchronizing automatically upon network restoration.

TABLE I: FUNCTIONAL TEST CASES AND VERIFICATION OUTCOMES

Test ID	Feature / Action	Expected Behavior	Observed Result
TC-01	User Auth & Login	JWT issued; profile synced	Passed (410 ms)
TC-02	GPS Tagging	Location fetched via GPS	Passed (<5 m error)
TC-03	Report Submission	Committed to Firestore	Passed (320 ms)
TC-04	Status Transition	Timeline steppers updated	Passed (100% sync)
TC-05	In-App Alerts	Notification pushed to user	Passed (<250 ms)
TC-06	Offline Fallback	Local cache activated	Passed (No crash)

VII. CONCLUSION

FixMyCity addresses longstanding inefficiencies in municipal infrastructure maintenance by providing an open, location-aware, and transparent civic engagement platform. By combining Android Jetpack Compose, Kotlin Coroutines, Firebase cloud services, and an administrative oversight portal, the system unifies the entire grievance lifecycle—from defect reporting and visual evidence capture to administrative triage and resolution verification. The platform's offline-tolerant repository architecture guarantees service continuity under varying connectivity conditions, while automated timeline tracking enhances mutual accountability between citizens and municipal authorities.

VIII. FUTURE SCOPE

Future enhancements include: (1) on-device machine learning (MobileNet) to automatically categorize defect severity and surface crack dimensions; (2) municipal GIS ward shapefile geofencing for automated complaint routing; (3) image hashing to detect and merge duplicate complaints; and (4) integration with municipal tax incentive credits to reward high-reputation civic contributors.

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